MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG3DSEPES200-FUNDAMENTALS OF KINESIOLOGY AND BIOMECHANICS

Duration 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer **All** Questions
Each question carries 1 mark

1. Who is considered a pioneer in the field of kinesiology?
   A. Aristotle B. Newton C. Galileo Leonardo da Vinci
   (Understanding 1)
2. What is the opposite of supination?
   A. Pronation B. Rotation C. Circumduction Abduction
   (Understanding 1)
3. Which of the following muscle is part of rotator cuff muscle?
   A. Deltoid B. Supraspinatus C. Pectoralis Major D. Latissimusdorsi
   (Understanding 1)
4. Which muscle group is responsible for hip flexion?
   A. Gluteus Group B. Hip Flexors C. Hamstrings D. Quadriceps(Understanding 4,5,6)
5. What is the unit of measurement for torque ?
   A. Newton (Nm) B Joules(J) Watts(W) D. Kilograms (kg)
   (Understanding 4,6)
6. What is the second law of motion ?
   A. Law of Inertia B. Law of acceleration C. Law of reaction D. Law of Gravity
   (Understanding 3,6)
7. Who formulated the laws of motion?
   B. A. Galileo Galilei B. Isaac Newton C. Albert Einstein D. Leonardo da Vinci
   (Understanding 4,6)
8. What is the primary goal of strength and conditioning programs?
   A. To improve flexibility B. To enhance cardiovascular endurance C. To increase muscular strength and power D. To improve body composition
   (Evaluate 5,7)
9. What is the importance o Fundamental Movement Screening (FMS) in sports?
   A. To assess athletic performance B. To identify potential injury risks C. To improve cardiovascular endurance D. To enhance muscular strength (Analyse 5)
10. What is the primary focus of the ''laws of motion and their biomechanical applications?
    A. Understanding human anatomy B .Applying physical laws to human movement
    C.Enhancing sports movements D. Preventing injuries (Apply 6)

[1 x 10 = 10]

Part B

Short Answer Type Questions
Answer **10** Questions
Each question carries **2** marks

11. Any two aims of kinesiology? (Understanding 1)
12. What is abduction and adduction (Apply 3,4)

13. What is the function of trapezius muscles in the upper extremity? (Remembering 4,5)
14. What is Rotator cuff muscles? (Remembering 4,5,6)
15. What is Rehabilitation (Understanding 4,5,6)
16. Define third law of motion? (Apply 6)
17. What is the role of biomechanics in injury prevention? (Analyse 5)

18. What is linear motion? (Remembering 4,6)
19. What is velocity? (Apply4,6)
20. Name any two benefits of strength training? (Understanding, 1)
21. Full form of FMS? (Apply,5)
22. Write the importance of Biomechanics? (Remembering 5)

[10 x 2 = 20]

Part C

Short Essay Type Questions

Answer **4** Questions

Each question carries **5**marks

23. Explain the importance of kinesiology in physical education and sports.? (Understanding 1)
24. Define pectoralis major and pectoralis minor. And its origin, insertion and actions.? (Analyse 3,4)
25. Explain 1st ,2nd and 3rd laws of motion and Its examples? (Apply 6)
26. Evaluate the role of biomechanics in injury prevention and performance enhancement in sports? (Remembering 3,6)
27. Write any 5 body movements and explain it? (Apply 3,4)
28. How does understanding biomechanics contribute to the development of effective strength and conditioning? (Understanding 4,5,6)

[4 x 5 = 20]